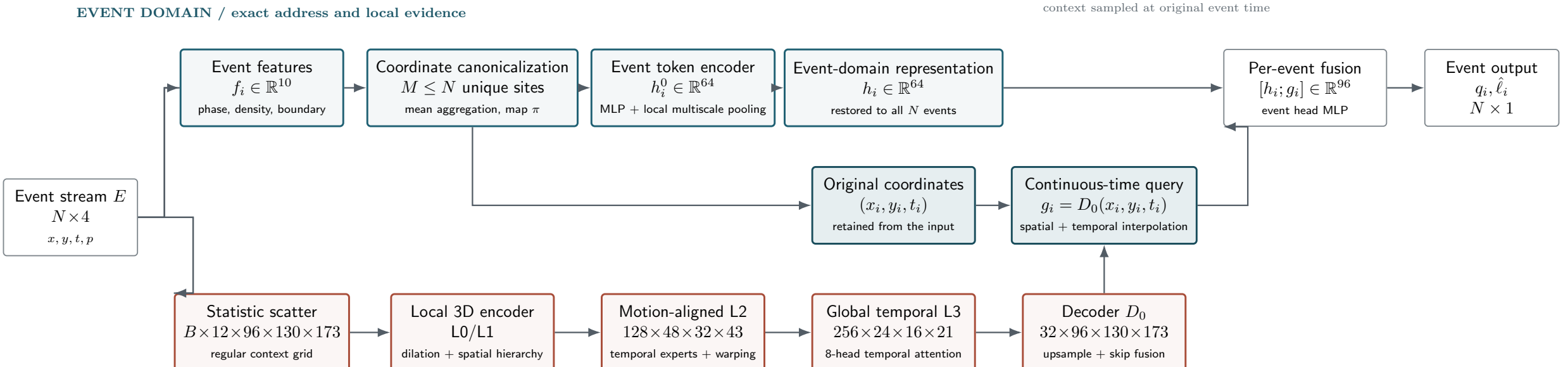

EVENT DOMAIN / exact address and local evidence
context sampled at original event time
Event features
$f_i \in \mathbb{R}^{10}$
phase, density, boundary
Coordinate canonicalization
$M \leq N$ unique sites
mean aggregation, map π
Event token encoder
$h_i^0 \in \mathbb{R}^{64}$
MLP + local multiscale pooling
Event-domain representation
$h_i \in \mathbb{R}^{64}$
restored to all N events
Per-event fusion
$[h_i; g_i] \in \mathbb{R}^{96}$
event head MLP
Event output
$q_i, \hat{\ell}_i$
$N \times 1$
Event stream E
$N \times 4$
x, y, t, p
Original coordinates
(x_i, y_i, t_i)
retained from the input
Continuous-time query
$g_i = D_0(x_i, y_i, t_i)$
spatial + temporal interpolation
Statistic scatter
$B \times 12 \times 96 \times 130 \times 173$
regular context grid
Local 3D encoder
L0/L1
dilation + spatial hierarchy
Motion-aligned L2
$128 \times 48 \times 32 \times 43$
temporal experts + warping
Global temporal L3
$256 \times 24 \times 16 \times 21$
8-head temporal attention
Decoder D_0
$32 \times 96 \times 130 \times 173$
upsample + skip fusion
FIELD DOMAIN / regularized spatiotemporal context